Chapter 7

TWO DIMENSIONAL MOTION (CARTESIAN)

7.1 Introduction :

Next in complexity to motion in a straight line is the two dimensional motion when the particle lies on a plane throughout the motion. An example of such a motion is the motion of a projectile i.e., the motion of a particle under gravity when it is projected with a velocity in a direction other than the direction of gravity. A second example is the motion of a particle under a central force i.e., a force always directed towards a fixed point (discussed in Chapter IX). In this chapter we could discuss some general ideas regarding two dimensional motion together with the detail discussion of a few simple types of motion in two dimensions.

7.2 Velocity and acceleration components in cartesian co-ordinates :

Let a particle while moving on a plane along the path APQ be at points P and Q at times t and $t + \Delta t$ respectively. With reference to a fixed rectangular axes OX and OY, let the co-ordinates of P be (x, y) and those of Q be $(x + \Delta x,\ y + \Delta y)$. The displacement PQ of the particle has components Δx and Δy parallel to OX and OY respectively. Hence the velocity component parallel to OX is

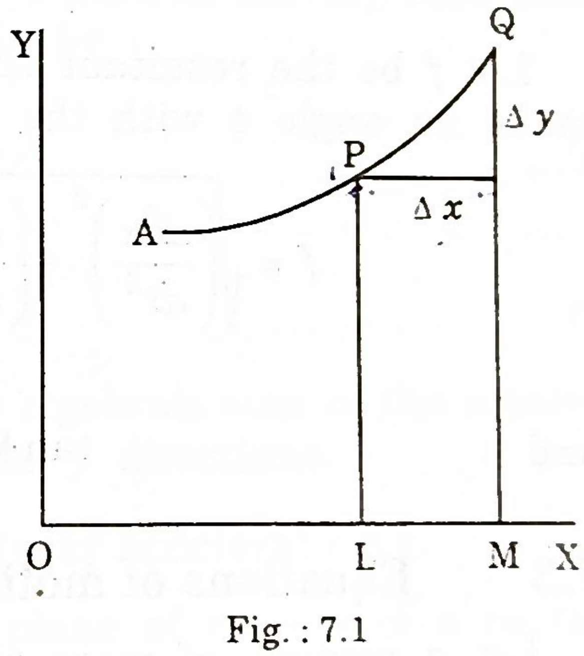


Fig. : 7.1

$$u_x \quad v_x = \underset{\Delta t \to 0}{\text{Lt}} \frac{\Delta x}{\Delta t} = \frac{dx}{dt} \qquad \text{...(1)}$$

Similarly, the velocity component parallel to OY is

$$v_y = \underset{\Delta t \to 0}{\text{Lt}} \frac{\Delta y}{\Delta t} = \frac{dy}{dt} \qquad \text{...(2)}$$

Let V be the resultant velocity of the particle at P at time t and let its direction make an angle θ with the positive direction of x-axis. Then

$$V = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} \quad \text{or,} \quad V = \sqrt{\dot{x}^2 + \dot{y}^2} \qquad \ldots(3)$$

and

$$\tan\theta = \frac{dy/dt}{dx/dt} = \frac{\dot{y}}{\dot{x}} \qquad \ldots(4)$$

Let u and v be the components of the velocity at P at time t parallel to the co-ordinate axes and those for the particle at Q at time $t + \Delta t$ are $u + \Delta u$ and $v + \Delta v$ respectively.

The acceleration f_x at P parallel to x-axis is

$$f_x = \lim_{\Delta t \to 0} \frac{(u + \Delta u) - u}{\Delta t} = \lim_{\Delta t \to 0} \frac{\Delta u}{\Delta t} = \frac{du}{dt}$$

$$\therefore \quad f_x = \frac{du}{dt} = \frac{d^2x}{dt^2} = \ddot{x} \qquad \ldots(5)$$

Similarly, the acceleration f_y at P parallel to y-axis is

$$f_y = \lim_{\Delta t \to 0} \frac{(v + \Delta v) - v}{\Delta t} = \lim_{\Delta t \to 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt}$$

$$\therefore \quad f_y = \frac{dv}{dt} = \frac{d^2y}{dt^2} = \ddot{y} \qquad \ldots(6)$$

Let f be the resultant acceleration at P and let its direction make an angle ϕ with the positive direction of x-axis. Then

$$f = \sqrt{\left(\frac{d^2x}{dt^2}\right)^2 + \left(\frac{d^2y}{dt^2}\right)^2} \quad \text{or,} \quad f = \sqrt{f_x^2 + f_y^2} \qquad \ldots(7)$$

and

$$\tan\phi = \frac{d^2y}{dt^2} \Big/ \frac{d^2x}{dt^2} \qquad \ldots(8)$$

7.3 Equations of motion :

Let a particle of mass m be moving along a plane curve C under the influence of a system of forces parallel to the co-ordinate axes. At any time t, let $P(x, y)$ be the position of the particle and X and Y are respectively the sum of the resolved parts of all the forces acting on the particle, parallel to the co-ordinate axes x and y. Then the equations of motion of the particle parallel to x and y-axes are respectively.

Let the inclination of the line joining the centre of the wheel be θ. The normal reactions of the rails on the wheels are R and S as shown in fig. 7.8. The vertical components of R and S balance the weight of the carriage while the horizontal components balance the centrifugal force Mv^2/r due to the motion of the carriage on the curved track. Thus

$$(R + S) \cos\theta = Mg$$

$$(R + S) \sin\theta = M\frac{v^2}{r}$$

$$\therefore \quad \tan\theta = \frac{v^2}{rg} \quad \text{or,} \quad \theta = \tan^{-1}\frac{v^2}{gr}$$

If h be the excess of the height of the outer rail over the inner one and d be the distance between the rails, then

$$h = d\tan\theta = \frac{v^2 d}{gr}$$

7.10 Expressions for the tangential and normal components of the acceleration of a particle describing a plane curve :

Let a particle be moving along a curve C, and P and P' be the positions of the particle at time t and $t + \Delta t$ respectively. Let the arc length $PP' = \Delta s$. The tangents at P and P' make angles

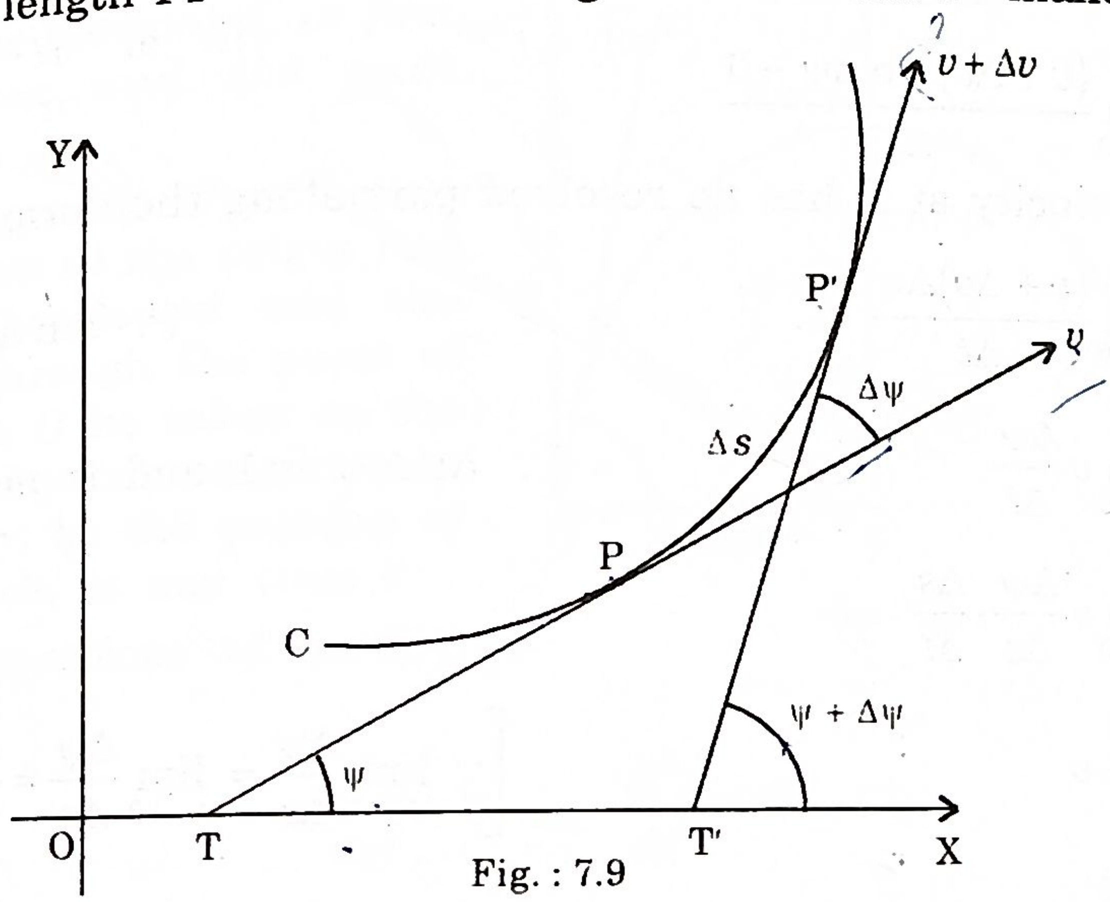


Fig. : 7.9

ψ and $\psi + \Delta\psi$ with a fixed axis OX. Let the velocity of the particle be v and $v + \Delta v$ along the tangents at P and P' respectively.

The acceleration along the tangent at P

$$f_t = \lim_{\Delta t \to 0} \frac{\text{Resolved part of } (v + \Delta v) \text{ along the tangent at } P - \text{the velocity along the tangent at } P}{\Delta t}$$

$$= \lim_{\Delta t \to 0} \frac{(v + \Delta v)\cos \Delta\psi - v}{\Delta t}$$

$$= \lim_{\Delta t \to 0} \frac{(v + \Delta v).1 - v}{\Delta t} \qquad [\because \cos \Delta\psi \simeq 1]$$

$$= \lim_{\Delta t \to 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt} = \frac{d^2 s}{dt^2} \qquad [\because v = ds/dt]$$

Alternatively,

$$f_t = \frac{dv}{dt} = \frac{dv}{ds} \cdot \frac{ds}{dt} = v\frac{dv}{ds}$$

Again, the acceleration along the normal at P

$$f_n = \lim_{\Delta t \to 0} \frac{\text{Resolved part of the velocity at time } (t + \Delta t) \text{ along the normal at } P - \text{the same at time } t}{\Delta t}$$

$$= \lim_{\Delta t \to 0} \frac{(v + \Delta v)\sin \Delta\psi - 0}{\Delta t}$$

(since velocity at P has no resolved part along the normal at P).

$$= \lim_{\Delta t \to 0} \frac{(v + \Delta v)\Delta\psi}{\Delta t} \qquad [\because \sin \Delta\psi \simeq \Delta\psi]$$

$$= \lim_{\Delta t \to 0} v\frac{\Delta\psi}{\Delta t} \qquad [\because \Delta v.\Delta\psi \ll 1 \text{ and is negligible}]$$

$$= \lim_{\Delta t \to 0} v\frac{\Delta\psi}{\Delta s} \cdot \frac{\Delta s}{\Delta t}$$

$$= v \cdot \frac{1}{\rho} \cdot v \qquad \left[\because \lim_{\Delta t \to 0} \frac{\Delta\psi}{\Delta s} = \lim_{\Delta s \to 0} \frac{\Delta\psi}{\Delta s} = \frac{d\psi}{ds} = \frac{1}{\rho}\right]$$

$$= v^2/\rho$$

Cor. If a particle moves along a circle of radius a, then we have $v = a\omega$, ω being the angular velocity of the particle and the radius of curvature $\rho = a$.

Hence, normal acceleration

$$= \frac{v^2}{\rho} = \frac{a^2\omega^2}{a} = a\omega^2$$

Thus, the resolved parts of acceleration along the tangent and normal (towards the centre) are respectively

$$\frac{dv}{dt} \text{ or } v\frac{dv}{ds} \text{ and } \frac{v^2}{a} \text{ or } a\omega^2$$

If v is constant $dv/dt = 0$ and the acceleration is along the normal only and is equal to v^2/a.

Further we can write $s = a\theta$ on a circle, with origin at the centre and θ being the angle made by the radius vector with the initial line. Thus

$$v = a\dot{\theta}$$

and the accelerations $a\ddot{\theta}$ and $a\dot{\theta}^2$ along tangential and normal directions.

7.11 Motion of a porjectile under gravity (supposed constant) :

A particle of mass m is projected into the air with velocity u in a direction making an angle α with the horizontal, to find its motion and the path described.

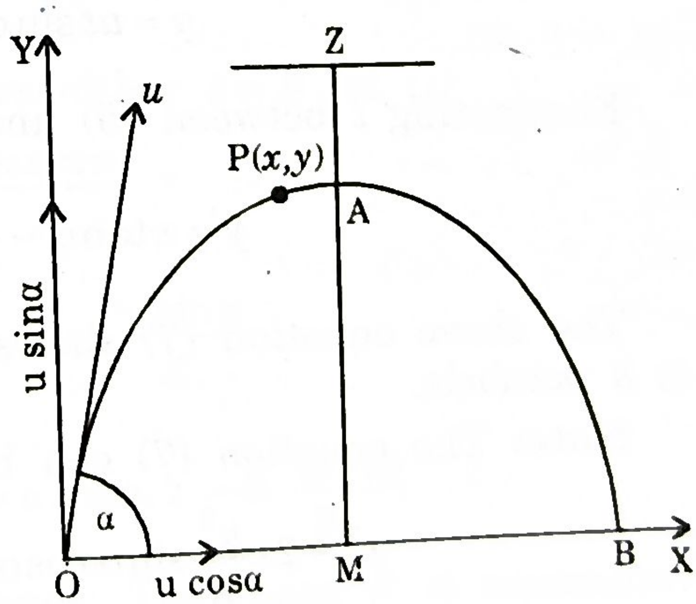


Fig. 7.10

Let the point of projection O be taken as the origin and let the horizontal and the vertical through the point of projection O be taken as the x-axis and y-axis respectively. Let $P(x, y)$ be the position of the particle at any time t.

The equations of motion are

$$m\frac{d^2x}{dt^2} = 0 \quad \text{or,} \quad \frac{d^2x}{dt^2} = 0 \qquad \ldots(1)$$

$$m\frac{d^2y}{dt^2} = -mg \quad \text{or,} \quad \frac{d^2y}{dt^2} = -g \qquad \ldots(2)$$

$| \; 2 \; \overline{m_1 + m_2}$

Dynamics

2D motion

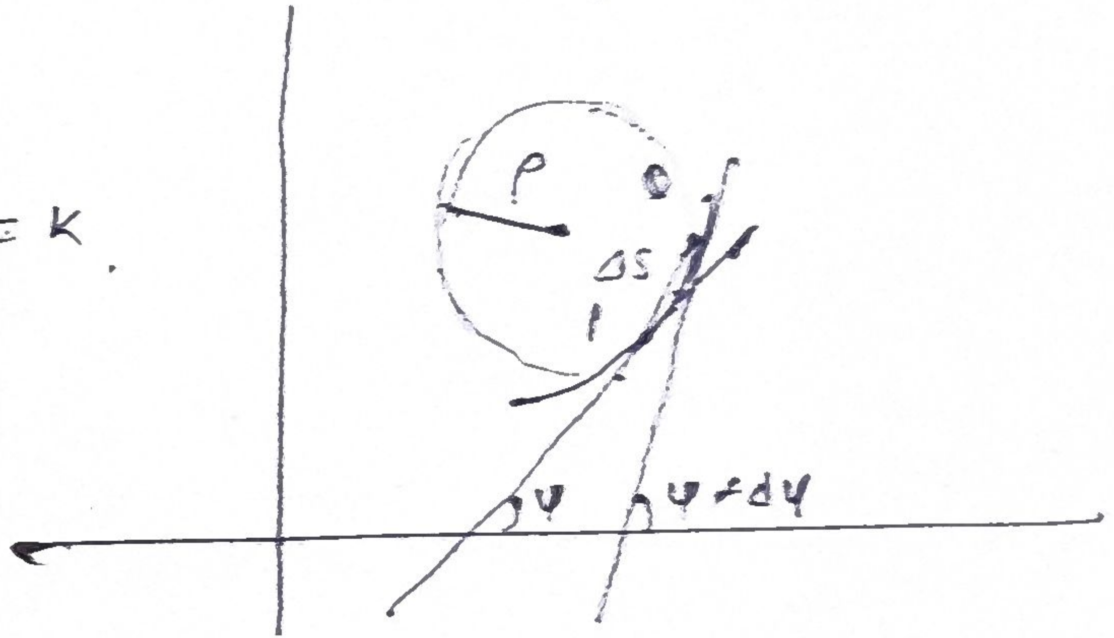
Cartesian & Polar

$\psi = f(s)$

$\theta \to 1 \quad \Delta s \to 0$

$$\lim_{\Delta s \to 0} \frac{\Delta \psi}{\Delta s} = \frac{d\psi}{ds} = K$$

$$\rho = \frac{1}{K} = \frac{ds}{d\psi}$$



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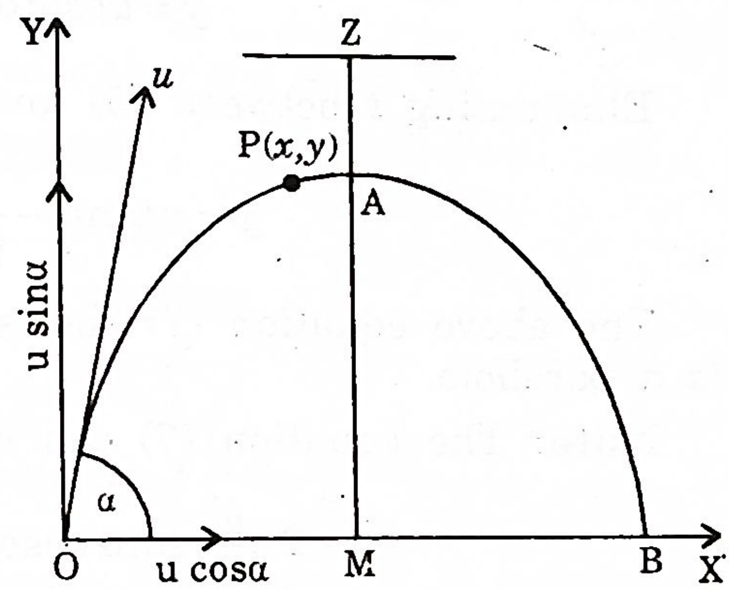

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The equations of motion are

$$m\frac{d^2x}{dt^2} = 0 \quad \text{or,} \quad \frac{d^2x}{dt^2} = 0 \qquad \ldots(1)$$

$$m\frac{d^2y}{dt^2} = -mg \quad \text{or,} \quad \frac{d^2y}{dt^2} = -g \qquad \ldots(2)$$

Integrating (1) and (2), we get

$$\frac{dx}{dt} = c_1 \text{ and } \frac{dy}{dt} = -gt + c_2$$

where c_1 and c_2 are constants.

Initially, when $t = 0$, $\frac{dx}{dt} = u\cos\alpha$ and $\frac{dy}{dt} = u\sin\alpha$.

$\therefore$ $c_1 = u\cos\alpha$ and $c_2 = u\sin\alpha$.

Hence,
$$\frac{dx}{dt} = u\cos\alpha \quad \ldots(3)$$

$$\frac{dy}{dt} = u\sin\alpha - gt \quad \ldots(4)$$

Again, integrating (3) and (4), we have

$$x = u\cos\alpha . t + c_3$$

$$y = u\sin\alpha . t - \frac{1}{2}gt^2 + c_4$$

Initially, $x = 0$, $y = 0$ when $t = 0$

$\therefore$ $c_3 = c_4 = 0$

Thus,
$$x = ut\cos\alpha \quad \ldots(5)$$

$$y = ut\sin\alpha - \frac{1}{2}gt^2 \quad \ldots(6)$$

Eliminating t between (5) and (6), we get

$$y = x\tan\alpha - \frac{gx^2}{2u^2\cos^2\alpha} \quad \ldots(7)$$

The above equation (7) shows that the path of the particle is a parabola.

Note: The equation (7) can be rewritten as

$$x^2 - 2x\frac{u^2}{g}\sin\alpha\cos\alpha = -\frac{2u^2\cos^2\alpha}{g}y$$

or,
$$\left(x - \frac{u^2}{g}\sin\alpha\cos\alpha\right)^2 = -\frac{2u^2\cos^2\alpha}{g}\left(y - \frac{u^2\sin^2\alpha}{2g}\right)$$

or,
$$\left(x - \frac{u^2\sin 2\alpha}{2g}\right)^2 = -\frac{2u^2\cos^2\alpha}{g}\left(y - \frac{u^2\sin^2\alpha}{2g}\right)$$

This represents a parabola with the following characteristics:

(i) The axis is vertical and vertex upwards.

(ii) The length of the latus rectum is $\dfrac{2u^2\cos^2\alpha}{g}$

(iii) The co-ordinates of the vertex are $\left(\dfrac{u^2\sin 2\alpha}{2g}, \dfrac{u^2\sin^2\alpha}{2g}\right)$.

(iv) The co-ordinates of the focus are $\left(\dfrac{u^2\sin 2\alpha}{2g}, -\dfrac{u^2\cos 2\alpha}{2g}\right)$.

(v) The height of the directrix $= MZ = AM + AZ$

=the height of the vertex + 1/4 × latus rectum

$$= \frac{u^2\sin^2\alpha}{2g} + \frac{u^2\cos^2\alpha}{2g} = \frac{u^2}{2g}.$$

(vi) The height of the focus above the point of projection =the height of the vertex −1/4 × latus rectum

$$= \frac{u^2\sin^2\alpha}{2g} - \frac{u^2\cos^2\alpha}{2g} = -\frac{u^2\cos 2\alpha}{2g}.$$

(vii) Time of flight : The time of flight is the time taken by the particle to reach the horizontal plane through the point of projection O.

Putting $y = 0$ in (6), we have either $t = 0$ (at O)

or,
$$T = \frac{2u\sin\alpha}{g}$$

Hence the time of flight is $T = \dfrac{2u\sin\alpha}{g}$

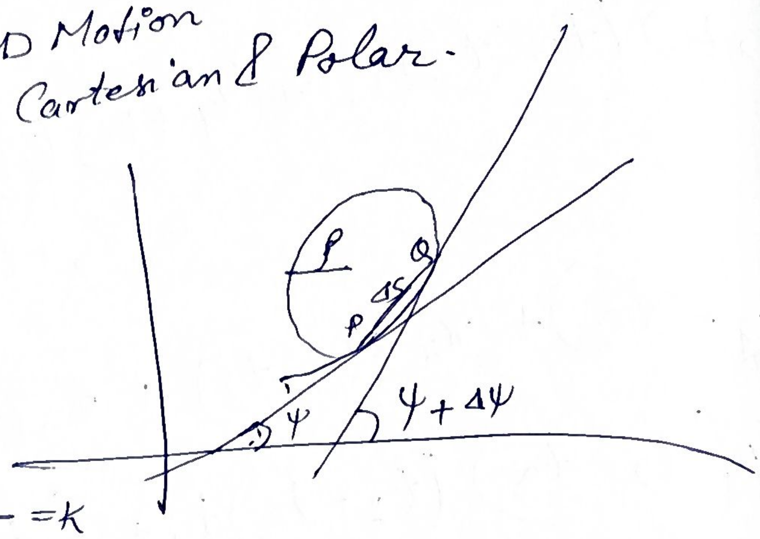
(viii) Horizontal range: $R = u\cos\alpha . T = \dfrac{u^2\sin 2\alpha}{g}$

(ix) Maximum horizontal range: The range R is maximum when $\sin 2\alpha = 1$, i.e., $\alpha = 45°$.

Hence the horizontal range is maximum when the angle of projection is 45°.

$\therefore$ Maximum horizontal range $= \dfrac{u^2}{g}$.

Dynamics : 2D Motion
Cartesian & Polar.



$\psi = f(s)$

$Q \to P \quad \Delta s \to 0$

$$\lim_{\Delta s \to 0} \frac{\Delta \psi}{\Delta s} = \frac{d\psi}{ds} = k$$

$$\rho = \frac{1}{k} = \frac{ds}{d\psi}$$

Motion of a projectile under gravity

$$y = x\tan\alpha - \frac{gx^2}{2u^2\cos^2\alpha}$$

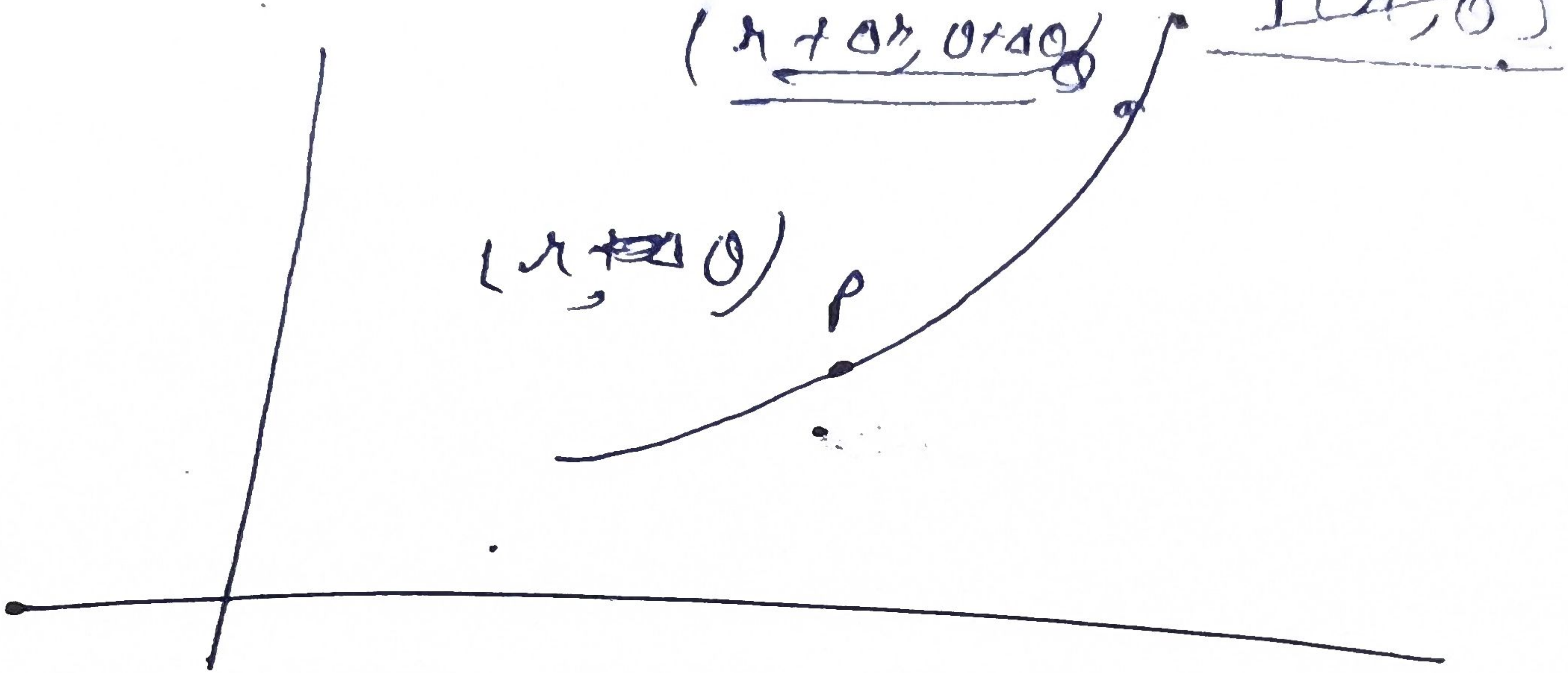
$$y = x\frac{\sin\alpha}{\cos\alpha} - \frac{gx^2}{2u^2\cos^2\alpha}$$

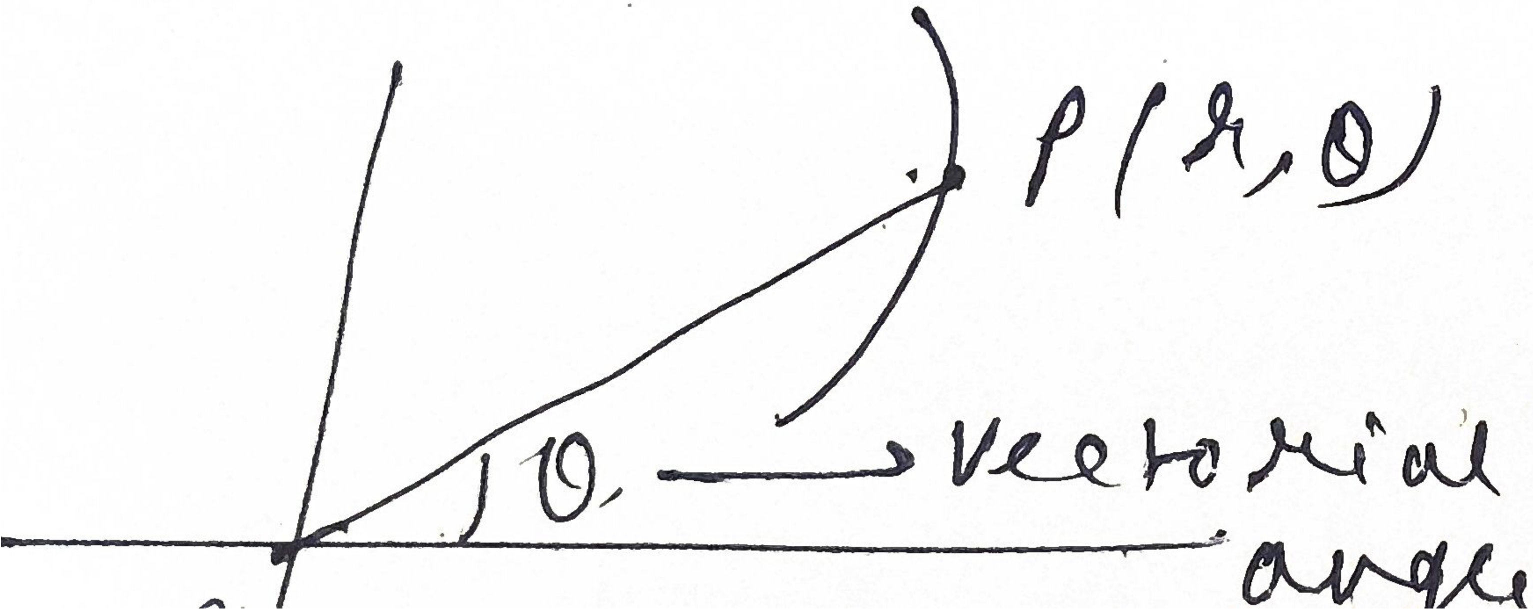
$$y\,2u^2\cos^2\alpha = \left(x\frac{\sin\alpha}{\cos\alpha} - \frac{gx^2}{2u^2\cos^2\alpha}\right)2u^2\cos^2\alpha$$

$$y\,2u^2\cos^2\alpha = \left(\frac{x\sin\alpha\,(2u^2\cos^2\alpha)}{\cos\alpha}\right) - gx^2$$

$$\frac{y\,2u^2\cos^2\alpha}{g} = \frac{x\sin\alpha\,2u^2\cos\alpha}{g} - x^2$$

$$-\frac{y\,2u^2\cos^2\alpha}{g} = x^2 - \frac{2xu^2\sin\alpha\cos\alpha}{g}$$





$OP = r$

= radius vector

Chapter 8

TWO DIMENSIONAL MOTION (POLAR)

8.1 Velocity and acceleration components in polar co-ordinates :

Let P and Q be the positions of a particle moving in a plane at times t and $t + \Delta t$ respectively. Let the polar co-ordinates of P and Q be (r, θ) and $(r + \Delta r, \theta + \Delta\theta)$ respectively with respect to a origin O and initial line OX. Thus $OP = r$, $\angle XOP = \theta$

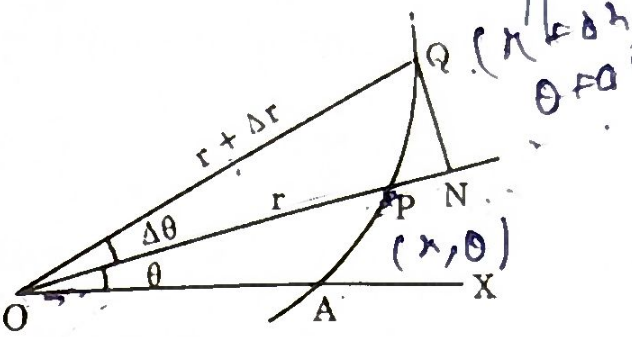


Fig. : 8.1

$$OQ = r + \Delta r, \ \angle XOQ = \theta + \Delta\theta$$

Draw QN perpendicular to OP. Let u, v be the components of velocity of the moving point along and perpendicular to the radius vector OP. By definition, we have

$$u = \underset{\Delta t \to 0}{\text{Lt}} \frac{\text{Displacement of the particle measured along } OP \text{ in time } \Delta t}{\Delta t}$$

$$= \underset{\Delta t \to 0}{\text{Lt}} \frac{ON - OP}{\Delta t} = \underset{\Delta t \to 0}{\text{Lt}} \frac{(r + \Delta r)\cos\Delta\theta - r}{\Delta t}$$

$$= \underset{\Delta t \to 0}{\text{Lt}} \frac{(r + \Delta r).1 - r}{\Delta t}$$

neglecting small quantities of second and higher orders

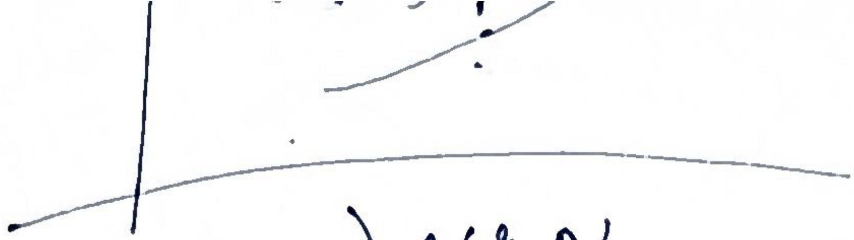
$$\therefore \quad u = dr/dt \qquad \ldots(1)$$

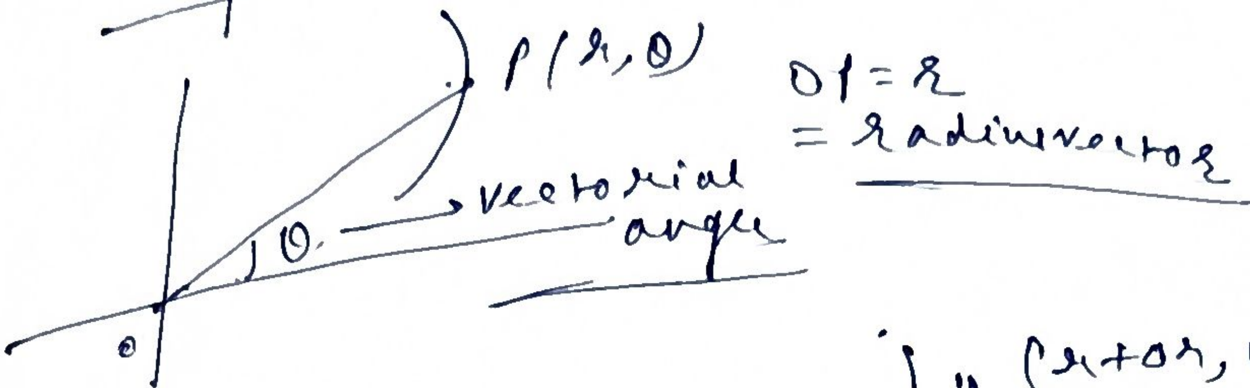
$$v = \underset{\Delta t \to 0}{\text{Lt}} \frac{\text{Displacement of the particle measured along perpendicular to } OP \text{ in time } \Delta t}{\Delta t}$$

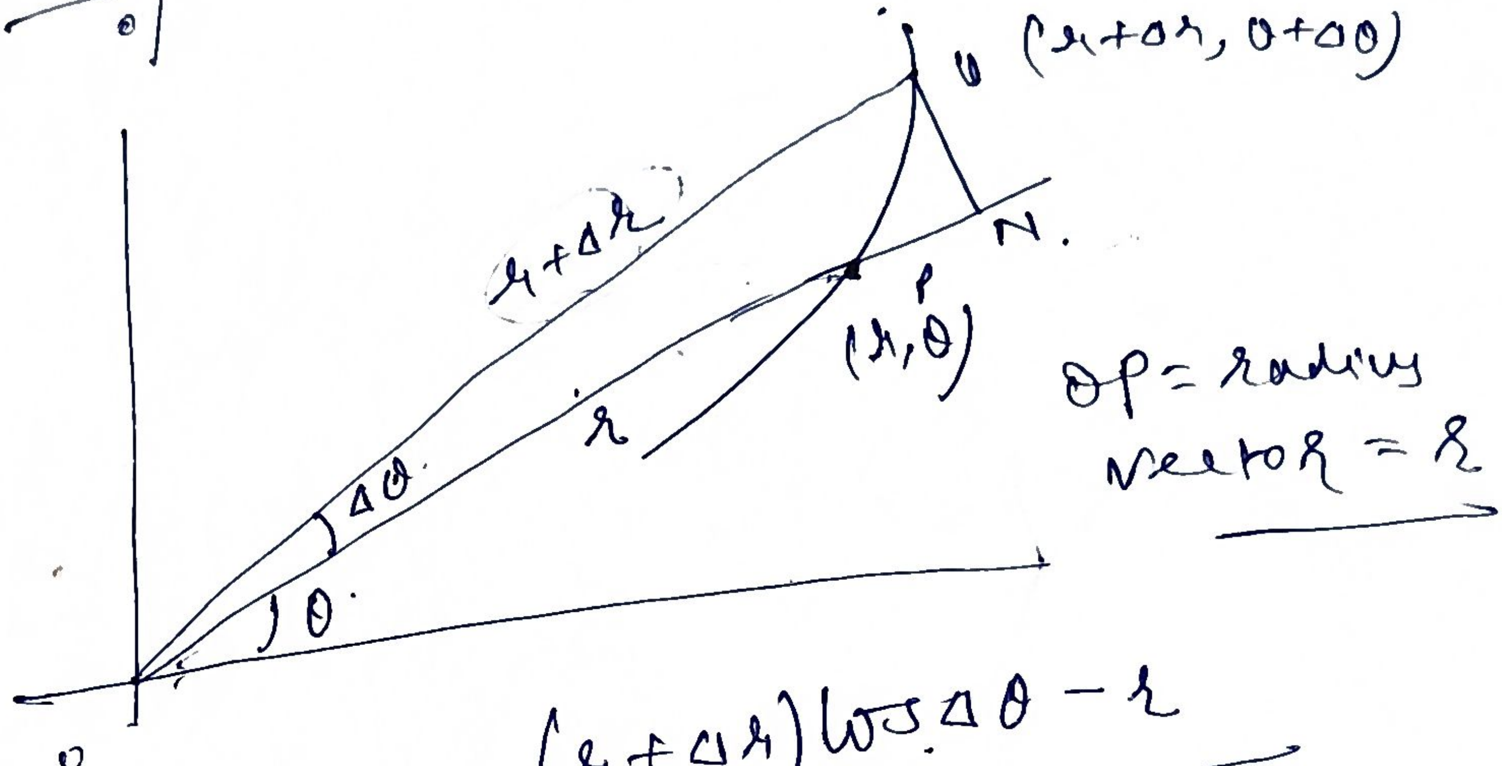
$$= \underset{\Delta t \to 0}{\text{Lt}} \frac{QN - 0}{\Delta t} = \underset{\Delta t \to 0}{\text{Lt}} \frac{(r + \Delta r)\sin\Delta\theta}{\Delta t}$$

$$= \underset{\Delta t \to 0}{\text{Lt}} \frac{(r + \Delta r)\Delta\theta}{\Delta t}$$

neglecting small quantities of second and higher orders.







$$v_r = \lim_{\Delta t \to 0} \frac{(r+\Delta r)\cos\Delta\theta - r}{\Delta t}$$

$$= \lim_{\Delta t \to 0} \frac{r + \Delta r - r}{\Delta t}$$

$$= \lim_{\Delta t \to 0} \frac{\Delta r}{\Delta t} = \frac{dr}{dt} = \dot{r}$$

$$v_\theta = \lim_{\Delta t \to 0} \frac{(r+\Delta r)\sin\Delta\theta - 0}{\Delta t}$$

$$= \lim_{\Delta t \to 0} \frac{(r+\Delta r)\Delta\theta}{\Delta t}$$

$$= \lim_{\Delta t \to 0} r\frac{\Delta\theta}{\Delta t} + \lim_{\Delta t \to 0} \frac{\Delta r\,\Delta\theta}{\Delta t}$$

$$= r\frac{d\theta}{dt} = r\dot{\theta}$$

$$\therefore \quad v = r d\theta/dt \qquad \ldots(2)$$

To compute the components of acceleration along the radial and cross-radial directions we assume the velocity components for the particle at P as u and v and those for the particle at Q as

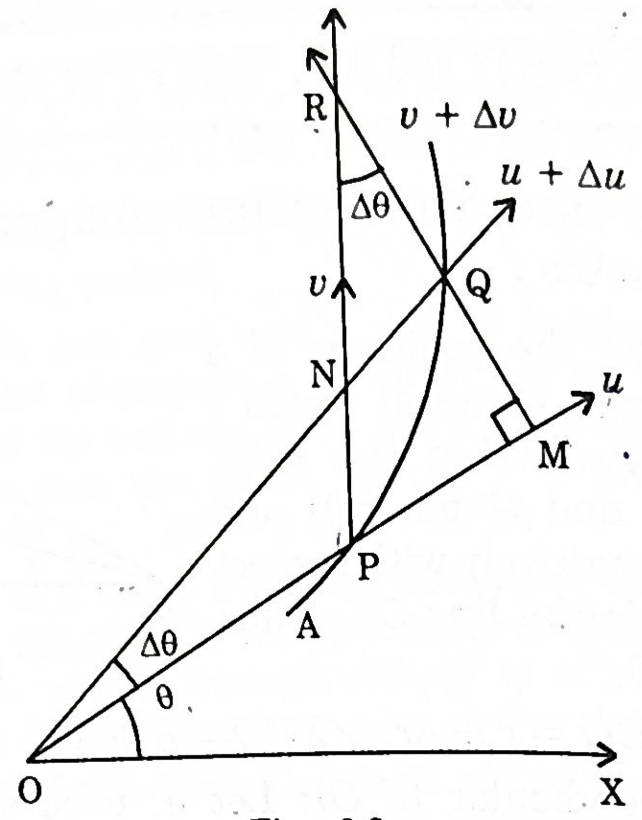


Fig. : 8.2

$u+\Delta u$ and $v+\Delta v$. Let the perpendicular to OQ at Q be produced to meet OP at M. Then the acceleration$(= f_r)$ of the moving point along OP is

$$f_r = \underset{\Delta t \to 0}{\text{Lt}} \frac{\begin{array}{c}\text{Velocity along } OP \text{ at time } (t+\Delta t) \\ -\text{its similar velocity at time } t\end{array}}{\Delta t}$$

$$= \underset{\Delta t \to 0}{\text{Lt}} \frac{\{(u+\Delta u)\cos\Delta\theta - (v+\Delta v)\sin\Delta\theta\} - u}{\Delta t}$$

$$= \underset{\Delta t \to 0}{\text{Lt}} \frac{(u+\Delta u).1 - (v+\Delta v)\Delta\theta - u}{\Delta t}$$

neglecting squares and higher powers of $\Delta\theta$

$$= \underset{\Delta t \to 0}{\text{Lt}} \frac{\Delta u - v\Delta\theta}{\Delta t}$$

neglecting $\Delta v \Delta\theta$ a second order small quantity.

$$\therefore \qquad f_r = \frac{du}{dt} - v\frac{d\theta}{dt}$$

Using (1) and (2), we have

$$f_r = \frac{d^2r}{dt^2} - r\left(\frac{d\theta}{dt}\right)^2 = \ddot{r} - r(\dot{\theta})^2 \qquad \ldots(4)$$

Similarly, the acceleration (f_θ) along the cross-radial direction is

$$f_\theta = \underset{\Delta t \to 0}{\text{Lt}} \frac{\{(u+\Delta u)\sin\Delta\theta + (v+\Delta v)\cos\Delta\theta\} - v}{\Delta t}$$

$$= \underset{\Delta t \to 0}{\text{Lt}} \frac{(u+\Delta u)\Delta\theta + (v+\Delta v).1 - v}{\Delta t}$$

$$= \underset{\Delta t \to 0}{\text{Lt}} \frac{u\Delta\theta + \Delta v}{\Delta t}$$

retaining only first order small quantities

$$\therefore \qquad f_\theta = u\frac{d\theta}{dt} + \frac{dv}{dt} \qquad \ldots(5)$$

Again, substituting from (1) and (2), we get

$$f_\theta = \frac{dr}{dt}\cdot\frac{d\theta}{dt} + \frac{d}{dt}\left(r\frac{d\theta}{dt}\right)$$

$$= 2\frac{dr}{dt}\frac{d\theta}{dt} + r\frac{d^2\theta}{dt^2}$$

or,
$$f_\theta = \frac{1}{r}\frac{d}{dt}\left(r^2\frac{d\theta}{dt}\right) \qquad \ldots(6)$$

Cor. For a particle moving in a circle of radius a we have the acceleration components along the radial and tangential directions to be $f_r = a\dot{\theta}^2$ and $f_\theta = a\ddot{\theta}$ since $r = a$, a constant $\dot{r} = 0$.

8.2 Alternative method (derivation from cartesian forms) :

Let the polar co-ordinates of $P(x, y)$ be (r, θ) with respect to O as the pole and OX as the initial line.

Here, we have

$$x = r\cos\theta \quad \text{and} \quad y = r\sin\theta \qquad \ldots(1)$$

Taking derivative of (1) with respect to t, we have

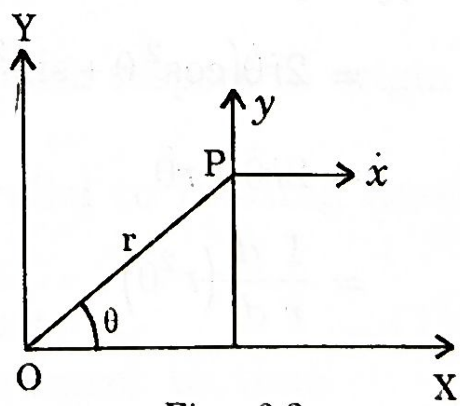


Fig. : 8.3